ASTANA IT UNIVERSITY · GROUP IOT-2603 · 2026

AuraLib

A smart university library: find a book and walk to its shelf, book a study room, borrow at self-checkout, and read more with friends.

Team: Zhubanysh · Sultan

500	10	9	13	20/20	3
titles on a 3-floor map	screens in the app	database tables, all with access rules	server functions enforce the rules	tests passed on the live database	languages: KZ · RU · EN

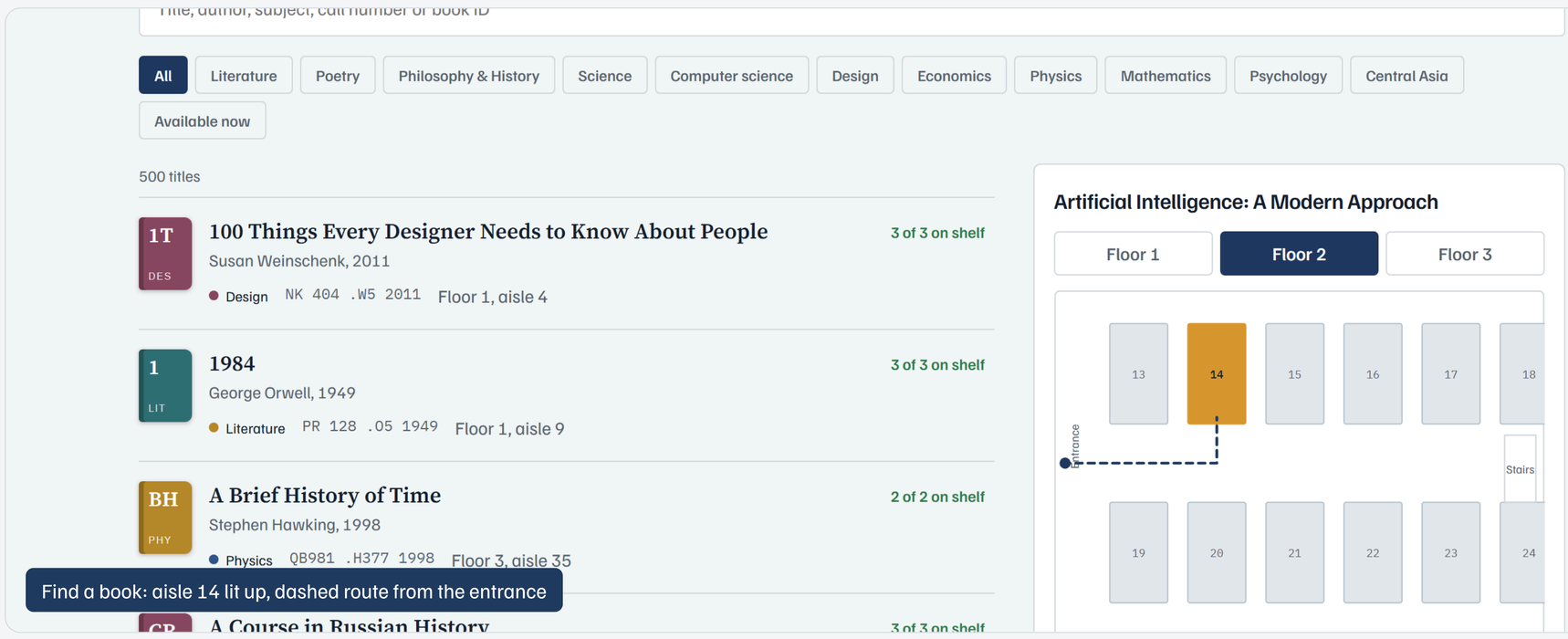
Try the live prototype
zhubanush.github.io/AituLib

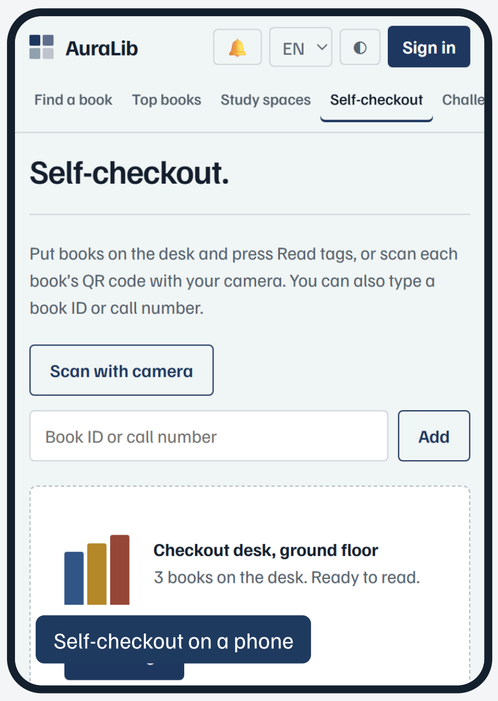
01 The problem

- **Where is the book?** The catalogue gives a call number, not the shelf. Students walk the floors or queue at the desk.
- **Who has the room?** Study rooms are booked on paper and in chats; the same hour is promised twice.
- **When is it back?** Every loan goes through the desk, and nobody knows when a popular book returns.

02 Our solution

- **Search + floor map:** the route from the entrance to the exact aisle and shelf.
- **Live room booking:** everyone sees the same grid; the database makes double booking impossible.
- **Self-checkout + hold queue:** scan a QR code, borrow for 21 days; the next reader is notified on return.





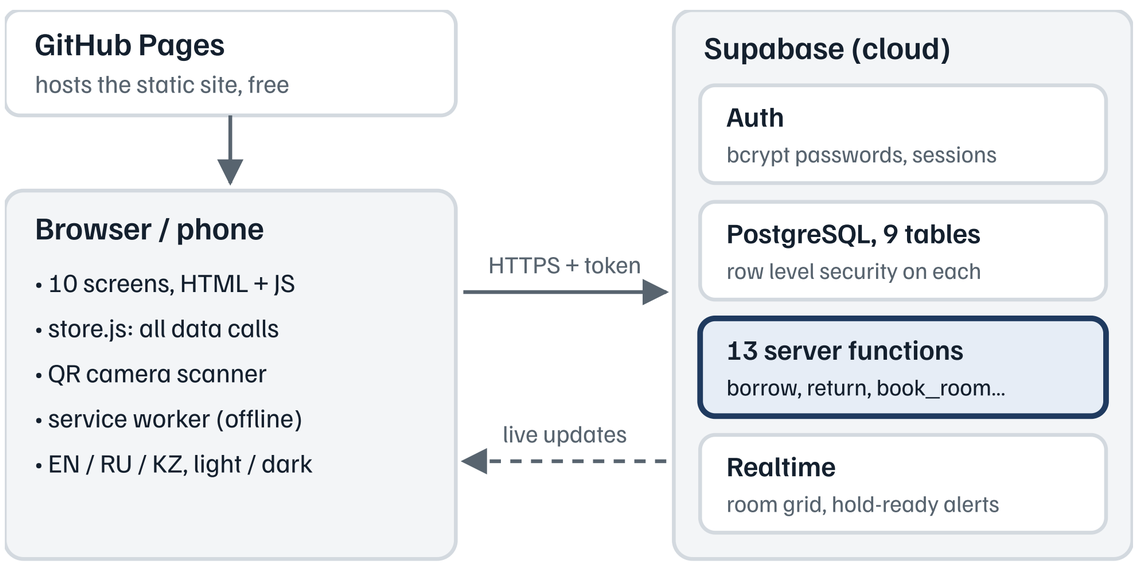
Reviews & ratings
only readers who borrowed a book

Reading challenge
monthly leaderboard, teams, 9 badges

Book club
vote for next week's book, discuss

Installable PWA
catalogue and map work offline

03 How it works



Every library rule runs on the server: two clicks on the same room-hour hit a unique constraint, so only one wins.

04 Smart planner

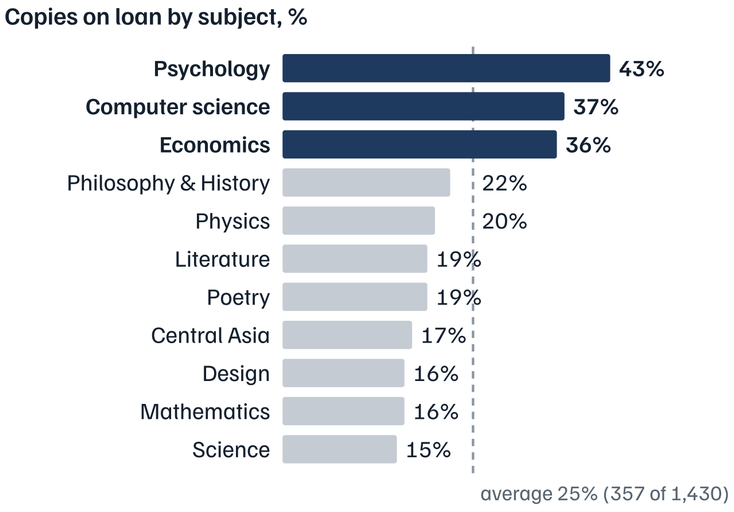
Input: group size, hours, day, time of day, equipment + live bookings. Every free room-hour is scored:

$$\text{score} = 100 - 5\cdot(\text{seats} - \text{people}) - 4\cdot\text{busy rooms/h} + 15\cdot\text{match} - 20\cdot\text{missing} \pm 10\cdot\text{time}$$

Input	Output
1 person, 2 h, Silent	**Quiet pod 2.A** — silent, right size
6 people, 2 h, Screen	**Room 3.05** — only room for 8
10 people	Warning: split the group
3 h, 2 h already booked	Warning: 1 h left today

Challenge points = 10 × books returned + 5 × different genres, each book once a month, max 3 a day.

05 Data: what students borrow



Finding 1. Psychology, CS and Economics have 36–43% of copies out, over twice Science (15%).

Finding 2. These 3 subjects hold 34% of the copies but 52% of the loans.

Decision. Buy extra copies for Psychology and CS first; do not expand Science or Design this semester.

06 Security

Risk	What we did
Exposed passwords hashes in the browser (A3)	Supabase Auth: bcrypt on the server, only a session token in the browser done
Broken access reading others' loans	Row level security on all 9 tables; changes only via checked functions done
XSS script in a review	All user text escaped done · Content Security Policy next
Fake accounts farming points	Rate limits, 3 books/day cap done · University SSO next

Next: catalogue in the database, librarian role, push alerts, a one-semester pilot on one library floor.

Prototype: **zhubanush.github.io/AituLib** · Code + database: **github.com/zhubanush/AituLib**

Catalogue availability before launch is simulated sample data.